



FOR JAVA DEVELOPERS MOVING TO NESTJS

385 MB to change one line.

Spring Boot to NestJS, episode 35: Docker for both stacks

An image is a build artifact with defaults of its own: what enters the build, what rebuilds, what reaches production, and who runs it.

With no `.dockerignore`, the Nest builder received 385 MB of context, and five macOS packages landed inside the Linux build stage.

A bare `node_modules` line only matches the top level: 40 MB were left. Matching it at any depth: 2 MB.

Both course images ran as root until a `USER` line changed it.

In scope

- `.dockerignore`
- Multi-stage builds
- Layer order and layered jars
- A non-root user

Not in scope

- Signals and graceful shutdown (episode 38)
- Image signing and scanning

Sizes are uncompressed local image sizes on this machine.



01 The measurements

NEST CONTEXT, NO .dockerignore	385 MB
NEST CONTEXT, **/node_modules	2 MB
NEST, ONE STAGE / MULTI-STAGE	1642 MB / 379 MB
SPRING, MAVEN IMAGE / JRE	707 MB / 283 MB
ONE LINE, COPY . . BEFORE npm ci	73 s
ONE LINE, PACKAGE FILES FIRST	13 s
SPRING FAT JAR / LAYERED	58.5 MB / 30.2 kB changed

Real builds, the build cache pruned first.

02 The Nest image

Ignore node_modules at any depth, install from the package files before copying source, ship only production dependencies, and run as the node user.

```
• • • .dockerignore
```

```
*/node_modules
dist
coverage
.git
*.log
*.tsbuildinfo
.env*
```

```
• • • Dockerfile
```

```
FROM node:24.21.0-alpine AS build
WORKDIR /app
COPY package*.json ./
RUN npm ci
COPY . .
RUN npm run build

FROM node:24.21.0-alpine
WORKDIR /app
COPY package*.json ./
RUN npm ci --omit=dev
COPY --from=build /app/dist ./dist
USER node
CMD ["node", "dist/main.js"]
```

after: uid 1000, GET /health 200

Run on a real machine, 2026-10-01, by scripts/verify-docker.sh in the course repository.



03 The Spring image

Extract the jar into Spring Boot's layers so a code change replaces the application layer only, and create a user to run as.

Dockerfile, the runtime stage

```
FROM eclipse-temurin:25-jre-alpine
RUN addgroup -S app && adduser -S app -G app
WORKDIR /app
COPY --from=build /app/extracted/dependencies/ ./
COPY --from=build /app/extracted/spring-boot-loader/ ./
COPY --from=build /app/extracted/snapshot-dependencies/ ./
COPY --from=build /app/extracted/application/ ./
USER app
ENTRYPOINT ["java", "org.springframework.boot.loader.launch.JarLauncher"]
```

one line changed: the application layer 30.2 kB, against 58.5 MB for the fat jar

Run on a real machine, 2026-10-01, by `scripts/verify-docker.sh` in the course repository.

Before you ship an image

1. What does the builder receive as context?
2. Do build tools or devDependencies reach the runtime image?
3. Does a one-line change reinstall dependencies?
4. Which layer changes when code changes?
5. Which user does the process run as?



Everything else

- | | |
|--|---|
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|  TikTok
https://www.tiktok.com/@codewithsamdev | Every measurement, one command: scripts/verify-docker.sh
https://github.com/code-with-sam-dev/spring-to-node |
|  Instagram
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https://docs.docker.com/build/concepts/context/ |

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